Module 3

Discrete-Time Dynamic Programming and Reinforcement Learning

In this document we present several MATLAB and Simulink homework problems and several engineering design projects relevant to dynamic programming and reinforcement learning in the discrete-time domain according to the material presented in Modules 1, 2, and 3.

MATLAB and Simulink Homework Problems

Homework Problem 3.1

Following Example 2.2 from Module 2, where the MATLAB live function for solving the liner-quadratic optimal control problem in *continuous-time* was considered, write a MATLAB code for a live function that solves the linear-quadratic optimal control problem in the *discrete-time* domain. Check controllability and observability by calling the corresponding live functions developed in Example 2.1 of Module 2. Note that the same controllability and observability tests apply to both continuous- and discrete-time domains.

Homework Problem 3.2

Write a code for a MATLAB live function that performs discretization of a linear system represented in the state space form with matrices A_c, B_c, C_c, D_c to obtain the corresponding discrete-time state space matrices A_d, B_d, C_d, D_d, for the given sampling period T_s. Using the obtained live function, demonstrate the code on the example of the continuous-time state space model of the islanded electric microgrid system whose state space matrices are given in Problem 2.9 of Module 2.

Homework **Problem 3.3**

Use the developed MATLAB live functions for examining system controllability and system observability developed in Example 2.1 of Module 2 to test controllability and observability of the discrete-time Boeing 747 aircraft model considered in Section 3.6 of Module 3.

Homework Problem 3.4

Build a Simulink block diagram to simulate dynamic behavior of the 14-th order computer communication network switch using the model data presented in Problem 2.4 from Module 2. Solve the corresponding LQ optimal discrete-time control problem with the quadratic performance penalty matrices being equal to identities.

Homework Problem 3.5

Write a code for a live function for examining stabilizability of a linear discrete-time dynamic system of order *n* represented by the state space matrices A and B. The system is stabilizable if the following conditions are satisfied for all unstable eigenvalues

$$\text{rank}\left[A-\lambda_i I \quad B\right]=n, \quad \forall i \quad \text{for which} \quad \left|\lambda_i(A)\right|>1, \quad i=1,2,...,n$$

In other words, the rank of the matrix $\left[A-\lambda_i I \quad B\right]=n$ is full for all unstable eigenvalue of matrix A. Demonstrate the live function on the example of the discretized model of an inverted pendulum on a cart from Section 3.5 of Module 3.

Homework Problem 3.6

Write a code for the live function that examines detectability of a linear discrete-time dynamic system of order *n* represented by the state space matrices A and C. The system is detectable if the following conditions are satisfied for all unstable eigenvalues

$$\text{rank}\begin{bmatrix} A-\lambda_i I \\ C \end{bmatrix}=n, \quad \forall i \quad \text{for which} \quad \left|\lambda_i(A)\right|>1, \quad i=1,2,...,n$$

In other words, the rank of the given matrix is full and equal to the system order for all unstable eigenvalue of matrix A.

Homework Problem 3.7

Write a MATLAB code for a live function that examines stability of a time-invariant discrete-time linear system. The live function should provide information about the number of unstable eigenvalues in an n-dimensional system. Demonstrate the live function on the example of an F-15 aircraft whose system matrix is given in Problem 2.11 in Module 2.

MATLAB and Simulink Reinforcement Learning Design Projects

In this section, we present several MATLAB discrete-time RL design projects, which should be implemented as closely as possible according to the MATLAB RL implementation structure presented in Figure 1.1 from Module 1 using MATLAB live scripts and live functions, eventually using the Simulink toolbox.

Design Project 3.1. RL Algorithm for the Electric Islanded Microgrid System

In this design project, students use MATLAB and its live scripts and live functions (and if needed Simulink) to demonstrate the RL solution (see block diagram in Figure 1.1 in Module 1) to the LQ optimal control problem in the discrete-time domain using the Hewer algorithm presented in Section 3.5 in Module 3.

Project Report Format: The project report is supposed to be typed, and it must contain all standard technical paper main parts: introduction (about the general methodology used), problem formulation (what you plan to do and what you are supposed to achieve) main results obtained (obtained simulation results, tables, and figures), discussion/comments (on obtained results and some observations), conclusions, appendix (with MATLAB or any other software code), references. The report should contain a brief description of the Hewer policy iteration algorithm with analytical presentation of all important formulas, equations, required conditions (assumptions).

Project Data: The project considers an islanded electric microgrid system whose continuous-time space matrices are presented in Problem 2.9 of Module 2. Discretize this system with the sampling rate $T_s = 0.01$. Assume that the penalty matrices in the corresponding LQ optimal control problem are identities, that is, given by $R_1 = I_4, R_2 = I_2$.

Project Task: Running policy iterations of the Hewer algorithm generate a table that shows how the approximate performance values converge per iteration to the optimal values. In general, you need to run only five to six iterations, in rare case nine or ten iterations due to the quadratic convergence of the algorithm. Plot on one figure all approximates feedback controls obtained for $k = 1, 2, 3, 4, 5$, and on another figure plot the approximate system state trajectories for a selected (the one that performs the worst) state variable for $k = 1, 2, 3, 4, 5$.

Hint: First check a simple MATLAB demo code done for an inverted pendulum presented in this project assignment. Use the MATLAB Live scripts with two sliders for the number of policy iterations and for the length of time interval of optimization.

Design Project 3.2. RL Algorithm for the Wind Turbine System

A continuous-time linear wind turbine mathematical model state space matrices are presented in Problem 2.10 in Module 2. Discretize this model with the sampling rate $T_s = 0.01$. To the

obtained discrete-time linear system apply the Hewer reinforcement learning algorithm following the project formulation and data presented in Problem 2.10 of Module 2 and Design Project 3.1 from Module 3. Write a live script code with corresponding live functions, including interactive involvement of Simulink whose role is to simulate the wind turbine dynamic response.

Design Project 3.3. RL Algorithm for a Satellite Optimal Control

Consider the satellite optimal control problem whose continuous-time state space matrices are given in Problem 2.10 of Module 2. Discretize this model with the sampling rate $T_s = 0.05$. Repeat all steps outlined of Design Project 3.1 from Module 3 for reinforcement learning optimal control of the satellite mathematical model. Write a live script code with corresponding live functions, including their interactive cooperation with Simulink whose role is to simulate the dynamic response of considered satellite model.

Design Project 3.4. RL Algorithm for an F-15 Aircraft Optimal Control

Consider the satellite optimal control problem whose continuous-time state space matrices are given in Problem 2.12 in Module 2. Discretize this model with the sampling rate $T_s = 0.05$. Repeat all steps outlined of Design Project 3.1 from Module 3 for RL optimal control of this aircraft. Implement the reinforcement learning mechanism according to Figure 1.1 from Module 1 using Simulink.

Design Project 3.5. RL Algorithm for a Drone Yaw Angle Optimal Control

Consider a drone yaw angle control continuous-time transfer function considered in Exercise 2.1 of Module 2. Obtain its state space form. Discretize the state space model obtained with the sampling rate $T_s = 0.01$. Repeat all steps outlined of Design Project 3.1 from Module 3 for RL optimal control drone yaw angle control. Implement the RL mechanism according to Figure 3.1 from Module 3 using Simulink.